

SIJIA LIAO

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EDUCATION

Doctor of Philosophy, University of Arizona

Statistics and Data Science, GPA: 4.0/4.0

Aug. 2020 - May 2026

Tucson, Arizona, USA

Master of Science, Sun Yat-Sen University

Probability and Mathematical Statistics, GPA: 4.0/4.0

Aug. 2017 - Jul. 2020

Guangzhou, Guangdong, China

Bachelor of Science, Sun Yat-Sen University

Mathematics and Applied Mathematics, GPA: 3.6/4.0

Aug. 2013 - Jul. 2017

Guangzhou, Guangdong, China

EXPERIENCE

Research Scientist Intern

TikTok Inc.

Jun. 2025 - Dec. 2025

Bellevue, Washington, USA

- Developed and productionized an indicator- and rationale-augmented training framework for a Vision Model to detect TikTok content violations, improving decision consistency/interpretability via rationale-based reasoning and delivering measurable F1 lift (+2-5%).
- Designed a dynamic EMA checkpoint ensembling algorithm to improve robustness across training snapshots; validated offline and adopted in the training pipeline, yielding +2-4% F1 and more stable performance on challenging slices.
- Performed hard-case mining and error analysis on model outputs to identify failure modes and drive model improvements.

Graduate Research Assistant/Associate

Mel and Enid Zuckerman College of Public Health, University of Arizona

Jan. 2023 - May 2025

Tucson, Arizona, USA

- Developed a **nonparametric** methodology for **high-dimensional data** analysis that integrates **variable selection and smoothing**, with rigorous support from theoretical analysis and simulation results.
- Developed an interpretable method for **scalar-on-image regression** models that captures spatially heterogeneous relationships (sparse and smooth) between the spatial predictors and scalar response.
- Designed an algorithm that achieves a 10% reduction on average in prediction error over P-splines and FPCR methods, while enhancing the identification rate of non-predictive regions by **90%**.

Graduate Teaching Assistant

Department of Mathematics, University of Arizona

Aug. 2021 - Dec. 2024

Tucson, Arizona, USA

- Aug. 2024 - Dec. 2024, Math/Stat 564 - Theory of Probability.
- Aug. 2022 - Dec. 2022, Math/Stat 263 - Introduction to Statistics and Biostatistics.
- Aug. 2021 - Dec. 2021, Jan. 2022 - May 2022, Math 112 - College Algebra Concepts and Applications.

PAPER

Interpretable Scalar-on-Image Linear Regression Models via the Generalized Dantzig Selector, Sijia Liao, Xiaoxiao Sun, Ning Hao, Hao Helen Zhang. Submitted, accepted by Journal of the American Statistical Association [[arXiv](#)] [[GitHub](#)].

Component Selection and Smoothing in High-Dimensional Additive Models, Sijia Liao, Hao Helen Zhang. Working paper.

HONOR AND AWARDS

Finalist, 2025 ASA Nonparametric Statistics Section Student Paper Awards

“Interpretable Scalar-on-Image Linear Regression Models via the Generalized Dantzig Selector.” Invited to present at the Joint Statistical Meetings (JSM) 2025, Nashville, Tennessee.

Herbert E. Carter Travel Award, FY2025 Session 3

University of Arizona Graduate College funding award for professional meetings, granted for traveling to JSM 2025, Nashville, Tennessee.

Honorable Mention, Outstanding Graduate Teaching Assistant (GTA) Awards

In recognition of the role as TA for the graduate level course Theory of Probability in the University of Arizona, Fall 2024.